Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-4 — Overview of Complex Numbers

Classwork — Give it a go

Show all your working. Replace i^2 by -1 at every step. A correct answer with no working is only half an answer.

Student name :	Date : .....

Part A — The imaginary unit and operations Questions 1 to 4

1. Write in the form bi with the radical simplified.

$$\sqrt{-98}$$

2. Evaluate. Show the remainder when the exponent is divided by 4.

$$i^{43}$$

3. Simplify. Take i out of each root first.

$$\sqrt{-6} \cdot \sqrt{-10}$$

4. Add and write the answer in the form $a + bi$.

$$(5 - 3i) + (-2 + 7i)$$

Part B — Multiply, conjugates and divide Questions 5 to 8

5. Multiply and write in the form $a+bi$.

$$(4-3i)(2+5i)$$

6. For $z=5-12i$ find $z\bar{z}$ and $|z|$.

$$z=5-12i$$

7. Divide. Multiply top and bottom by the conjugate.

$$\frac{6+2i}{1+3i}$$

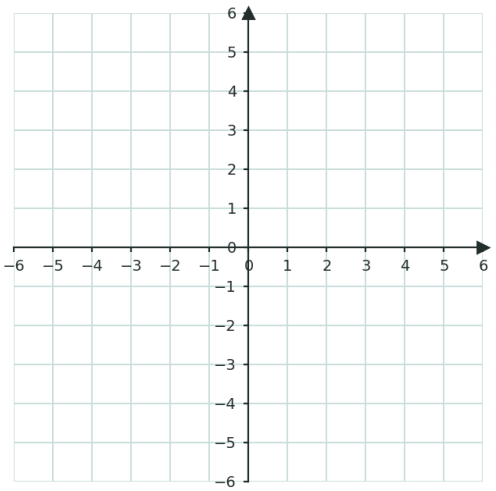
8. Solve. Write both solutions in the form $a+bi$.

$$x^2-6x+13=0$$

Part C — The plane, a factorisation and an error Questions 9 to 11

9. Plot $z = -3 + 2i$ and its conjugate on the complex plane (horizontal axis real, vertical axis imaginary). Label each point.

$$z = -3 + 2i, \quad \bar{z} = -3 - 2i$$



10. Factor over the complex numbers.

$$x^2 + 25$$

11. A student writes this. Say what is wrong, then give the correct value.

$$\sqrt{-4} \cdot \sqrt{-9} = \sqrt{36} = 6$$